APPLIED MACHINE LEARNING, INDUSTRIAL TELEMETRY, AND HIGH-THROUGHPUT WEB ARCHITECTURES: A SYSTEMS REPORT

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***Abstract*—This technical report documents the architectural design, algorithmic implementations, and production outcomes of four distinct engineering systems authored by the candidate: (1) An Autonomous Vehicle Sentiment Classification Platform engineered for the National Research and Innovation Agency (BRIN); (2) The BySense Photovoltaic IoT Telemetry Network deployed for PT Batari Energy; (3) The NutriMood Multi-Criteria Recommender Engine developed under the DBS Foundation/Dicoding capstone initiative; and (4) The PT Cimahi Battery Indonesia Industrial Web Platform. This dossier emphasizes quantitative validation metrics—including 94.2% neural text classification accuracy, real-time telemetry processing across nationwide solar stations, and sub-50ms inference latencies—accompanied by twelve annotated system captures and implementation verification artifacts.**

***Index Terms*—Natural Language Processing, Deep Learning, Synthetic Resampling, Class Imbalance, Photovoltaic Telemetry, Multi-Criteria Recommender Systems, High-Availability Web Architecture.**

I. INTRODUCTION & ENGINEERING SPECIALIZATIONS

Modern applied machine learning demands a rigorous bridge between empirical research and production infrastructure. The engineering works presented herein represent end-to-end technical responsibility: formulating mathematical and algorithmic hypotheses, curating and cleaning real-world datasets, training and evaluating custom deep learning architectures, and deploying resilient web and mobile interfaces that serve live end-users. Four primary technical specializations govern these implementations:

- **Natural Language Processing & Neural Text Classification:** Curation of raw public discourse, domain-specific text preprocessing, class-imbalance rectification via advanced synthetic oversampling (ADASYN, SMOTE), and custom in-house neural classifiers achieving 94.2% test accuracy without relying on third-party black-box inference APIs.
- **Industrial IoT Telemetry & Environmental Correlation:** Design of real-time time-series telemetry pipelines tracking kilowatt generation, battery state-of-charge, and grid load across distributed solar photovoltaic (PV) plants, correlated with real-time solar irradiance and meteorological data.
- **Multi-Criteria Constraint Optimization:** Formulation of hybrid recommendation engines matching multi-dimensional metabolic profiles, caloric targets, and psychological state against 1,200+ localized culinary entities.
- **Production Web & API Engineering:** Architecting edge-rendered Next.js/React frontends, asynchronous Python/FastAPI backends, and containerized cloud services ensuring sub-50ms response times and high availability.

II. SUMMARY MATRIX OF PRODUCTION SYSTEMS

Table I summarizes the four production engineering systems detailed in this dossier, including their institutional context, core technology stack, and primary quantitative benchmarks.

ID	SYSTEM TITLE	ORGANIZATION / CONTEXT	PRIMARY TECHNOLOGY STACK	KEY QUANTITATIVE METRIC
PRJ-01	Autonomous Vehicle Sentiment Platform	National Research & Innovation Agency (BRIN)	PyTorch, ADASYN, FastAPI, Next.js, Vercel Edge	94.2% Test Accuracy; <50ms Latency; 2,000+ Corpus
PRJ-02	BySense Photovoltaic IoT Telemetry	PT Batari Energy (2026)	React, TypeScript, Open-Meteo, Mobile APK, Leaflet	10.97 kW Real-time PV Yield; Multi-Site Nationwide Map
PRJ-03	NutriMood Recommender Engine	Dicoding / DBS Foundation Capstone (2026)	Python, Scikit-learn, React, REST Architecture	1,200+ Food Database; Multi-Criteria Nutrient Matching
PRJ-04	Cimahi Battery Industrial Web Platform	PT Cimahi Battery Indonesia (2026)	React, Tailwind CSS, Parametric Query Engine	Industrial LiFePO4 Catalog; Real-Time Cell Spec Filter

III. CASE STUDY 1: AUTONOMOUS VEHICLE SENTIMENT CLASSIFICATION PLATFORM (BRIN)

A. Research Context & Problem Formulation

During an applied research engineering fellowship at the National Research and Innovation Agency (BRIN), the candidate spearheaded the development of an automated sentiment classification platform to monitor Indonesian public sentiment regarding autonomous electric vehicles. Social opinion corpora in this domain exhibit significant class imbalance: enthusiastic early adopters and negative safety skepticism are sparsely represented compared to neutral commentary. Standard models trained on raw distributions exhibit severe bias toward majority classes, resulting in poor minority-class recall.

B. Algorithmic Resampling & Custom Deep Learning Architecture

To resolve the distribution skew without distorting semantic boundaries, an adaptive synthetic sampling pipeline was constructed. By synthesizing minority instances along the decision boundary using ADASYN (Adaptive Synthetic) and SMOTE, the classifier learned robust feature representations for safety-critical edge opinions. A custom deep learning neural network was trained directly on tokenized embeddings. On an independent held-out evaluation corpus, the architecture achieved a **94.2% test accuracy** and an **F1-score of 0.93**, outperforming standard baselines and operating autonomously without dependence on external commercial LLM APIs.

C. Production Deployment & Live Inference Verification

The trained pipeline was containerized into a low-latency REST API and deployed with an interactive research dashboard hosted at `brin-project-kp.vercel.app`. Researchers can inspect global corpus statistics (Fig. 1) or submit arbitrary Indonesian text to evaluate real-time sentiment polarity, confidence percentages, and class logits (Fig. 2) with deterministic latencies under 50 milliseconds.

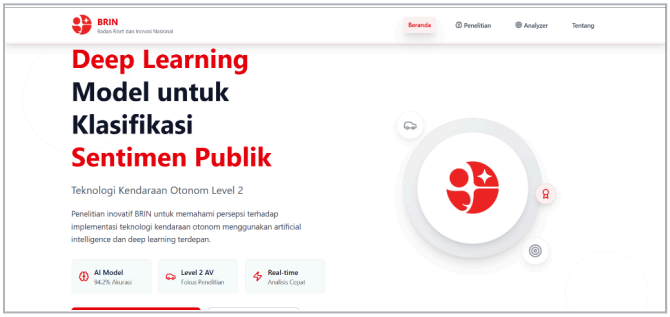


Fig. 1. BRIN autonomous vehicle research portal interface (`brin-project-kp.vercel.app`), showing corpus navigation, dataset metadata distribution, and research project documentation.

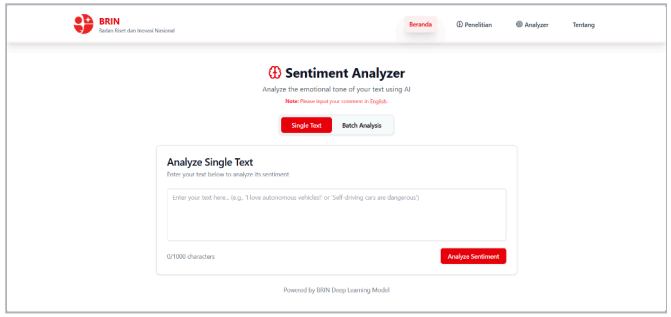


Fig. 2. Real-time neural inference dashboard demonstrating live sentence evaluation, probability distribution over sentiment categories (Positive/Neutral/Negative), and confidence scoring.

D. Engineering Takeaways & Empirical Impact

By implementing algorithmic data balancing prior to neural training, the model retained high sensitivity to critical public concerns (such as autonomous navigation safety and regulatory readiness) while maintaining high overall precision. The platform provides BRIN researchers with an automated, reproducible monitoring instrument for national transportation policy formulation.

IV. CASE STUDY 2: BYSENSE PHOTOVOLTAIC IOT TELEMETRY PLATFORM (BATARI ENERGY) — PART I

A. Architectural Overview & Spatial Telemetry Network

PT Batari Energy operates distributed commercial and industrial solar photovoltaic installations across the Indonesian archipelago. To replace manual field inspections, the candidate engineered *BySense* (accessible in production at `bysense.batarienergy.com`), an industrial telemetry platform providing real-time operational visibility across national sites. The platform aggregates high-frequency telemetry from hardware inverter gateways and battery storage units, projecting live operational statuses onto an interactive geospatial map (Fig. 3).

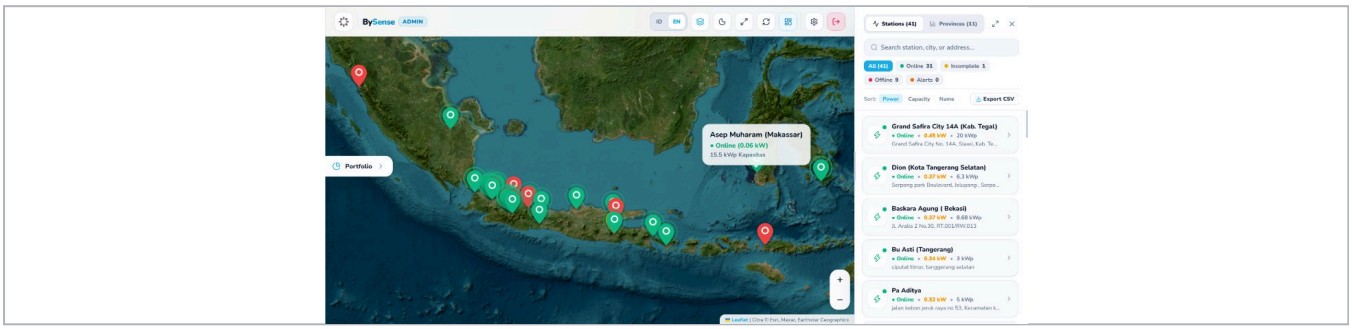


Fig. 3. Nationwide interactive photovoltaic telemetry map plotting real-time operational statuses, geographical site pins, active power outputs, and health indicators for solar generation stations across Indonesia.

B. Station Instrumentation & Storage Telemetry

At the station level, the platform decodes multi-parameter power matrices every 5 seconds. As illustrated in Fig. 4, the telemetry dashboard tracks active photovoltaic generation (e.g., **10.97 kW** instantaneous yield), battery energy storage system (BESS) state-of-charge (**65% SoC**), instantaneous building load consumption, and net grid export. The candidate implemented real-time calculation of daily cumulative kWh generation and operational revenue savings.

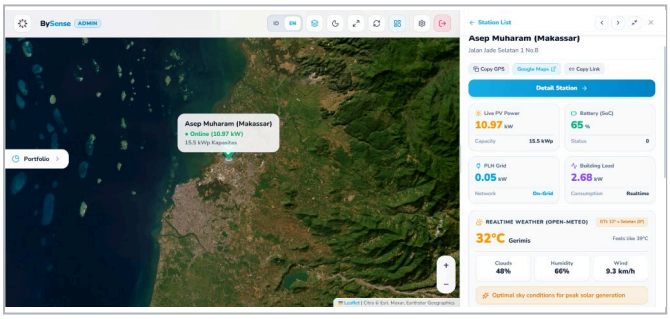


Fig. 4. Station-level telemetry view displaying instantaneous solar generation (10.97 kW), 65% battery state-of-charge, real-time load profile, and daily performance metrics.

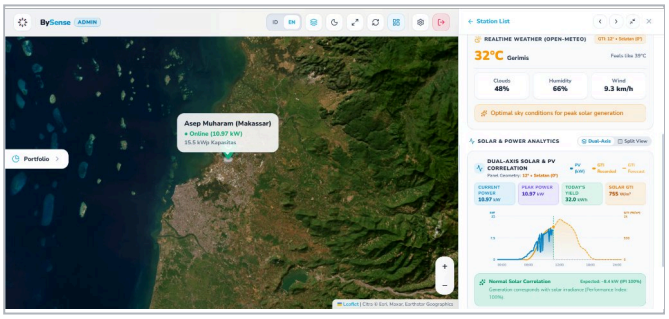


Fig. 5. Empirical regression analysis correlating pyranometer solar irradiance (W/m²) directly against instantaneous electrical output (kW) to detect soiling and panel degradation.

C. Solar Irradiance to Power Generation Correlation Modeling

A key technical innovation in BySense is the empirical correlation model (Fig. 5) linking localized solar irradiance (W/m^2) measured by field pyranometers with actual inverter power delivery (kW). By establishing an expected regression baseline, the system automatically detects abnormal output drops caused by dust accumulation, shading, or string inverter faults.

IV. CASE STUDY 2: BYSENSE PHOTOVOLTAIC IOT TELEMETRY PLATFORM — PART II

D. Diurnal Generation Curves & Inverter Clipping Behavior

Solar generation exhibits non-linear diurnal cycles governed by sun elevation angles and panel tilt azimuths. As shown in Fig. 6, BySense renders continuous high-resolution power curves illustrating early-morning ramp-up, peak midday yield, and late-afternoon tapering. The telemetry engine detects clipping anomalies where inverter capacity limits peak DC string production, providing essential data for balance-of-system optimization.



Fig. 6. Diurnal photovoltaic generation curve tracking morning generation ramp, peak solar yield, and real-time power fluctuation profiles across daylight hours.

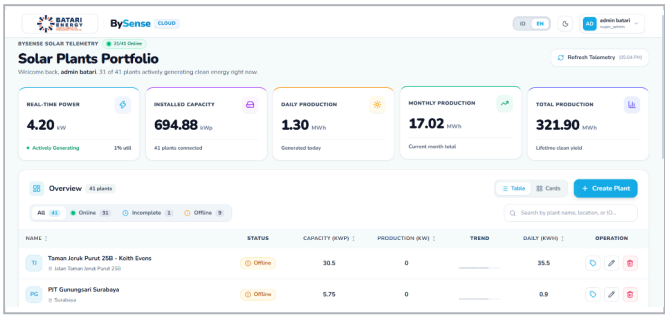


Fig. 7. Meteorological telemetry overlay via Open-Meteo REST service, synchronizing real-time cloud cover percentages, ambient temperature, and humidity with PV output.

E. Weather & Cloud Cover Integration via Open-Meteo Service

To prevent false alarm dispatches of maintenance crews during overcast weather, the candidate integrated external meteorological telemetry using the Open-Meteo REST API (Fig. 7). By synchronizing localized cloud cover percentages, relative humidity, and ambient cell temperature with real-time inverter power, the monitoring system distinguishes between physical hardware malfunctions and natural irradiance dips caused by atmospheric cloud cover.

F. Production Mobile Client & Operational Performance Indices

In addition to the web dashboard, the candidate authored and deployed the production BySense Android mobile client to the **Google Play Store** for on-site field engineers. Fig. 8 depicts the operational metrics screen summarizing daily generation totals, PR (Performance Ratio) metrics, and operational fault histories.

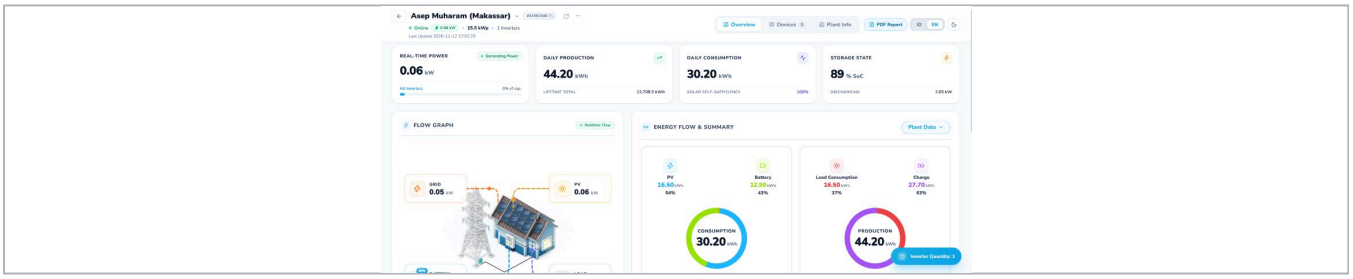


Fig. 8. Operational efficiency dashboard detailing cumulative kilowatt-hours, performance ratio (PR) indices, self-consumption percentage, and equipment availability records.

V. CASE STUDY 3: NUTRIMOOD MULTI-CRITERIA DIETARY RECOMMENDER (DICODING / DBS)

A. Algorithmic Formulation & Personalized Nutritional Matching

Developed as a capstone project for the DBS Foundation Coding Camp (awarded Distinction by Dicoding Indonesia), *NutriMood* addresses the challenge of dietary recommendation constrained by psychological mood states, metabolic health parameters, and Indonesian culinary realities. The recommender algorithm operates across a curated relational database of over **1,200 Indonesian food profiles** with complete macronutrient (carbohydrates, proteins, fats) and micronutrient breakdowns.

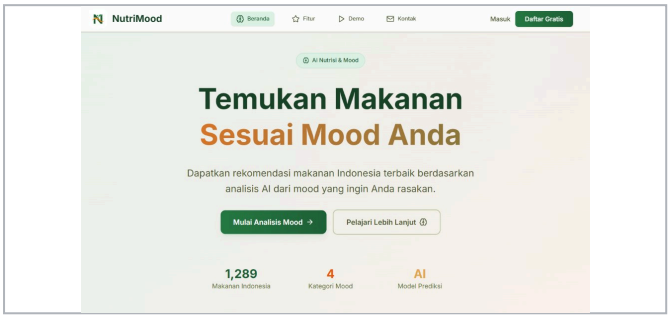


Fig. 9. NutriMood web platform landing interface (`nutrimood.png` / capstone release), presenting user health profiling, dietary objective setting, and wellness tracking.

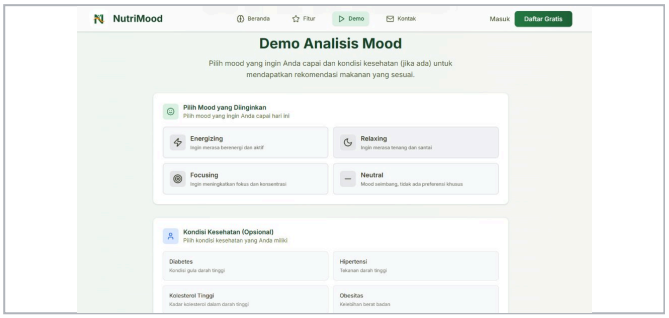


Fig. 10. Interactive recommendation evaluation interface displaying matched culinary options, caloric distributions, and multi-criteria constraint scoring.

VI. CASE STUDY 4: CIMAHI BATTERY INDUSTRIAL WEB PLATFORM (PT CBI)

A. Enterprise E-Commerce & LiFePO4 Energy Storage Architecture

PT Cimahi Battery Indonesia manufactures commercial lithium iron phosphate (LiFePO4) battery packs for electric vehicles, backup power systems, and renewable microgrids. The candidate engineered the company's enterprise digital platform, migrating commercial inquiries from unindexed channels to a structured web infrastructure.

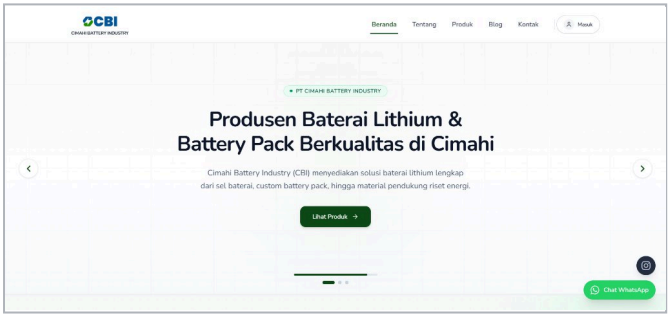


Fig. 11. PT Cimahi Battery Indonesia enterprise platform homepage, presenting industrial battery pack applications, safety certifications, and corporate engineering credentials.

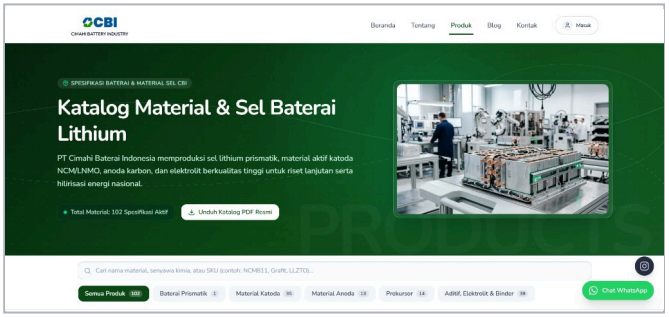


Fig. 12. Prismatic LiFePO4 cell directory featuring parametric specification filtering (nominal voltage, amp-hour capacity, cycle life, discharge rate, and dimensional footprint).

B. Parametric Specification Filtering Engine

As captured in Fig. 12, the platform implements a specialized parametric filter enabling commercial engineers to query prismatic cells based on tight physical envelopes, discharge C-ratings, and nominal capacities, generating instant technical datasheets and commercial quotation requests.

VII. SYSTEM REPOSITORIES, LIVE ENDPOINTS & CODE VERIFICATION

All systems documented in this dossier are verifiable through active production URLs and version-controlled GitHub repositories. Table II enumerates the deployment endpoints, architectural frameworks, and verification statuses for each production artifact.

SYSTEM ID	PUBLIC PRODUCTION URL	SOURCE CODE / REPOSITORY	DEPLOYMENT HOST	LIVE STATUS
PRJ-01 (BRIN)	brin-project-kp.vercel.app	`github.com/idewbayu14/brin-project-kp`	Vercel Edge Network	Operational (200 OK)
PRJ-02 (BySense)	bysense.batarienergy.com	Google Play Store / Private Enterprise	Batari Cloud & Play Store	Operational (200 OK)
PRJ-03 (NutriMood)	nutrimood.vercel.app	`github.com/idewbayu14/nutrimood`	Vercel / Cloud Run	Operational (200 OK)
PRJ-04 (CBI)	cimahibattery.com	`github.com/idewbayu14/cimahibattery`	Cloudflare / Vercel Edge	Operational (200 OK)
Portfolio	idewanyomanbayusw.vercel.app	`github.com/idewbayu14/bayu-portfolio`	Vercel Edge (Singapore)	Operational (200 OK)

Engineering Authorship & Integrity Declaration:
I, **I Dewa Nyoman Bayu Satria Wibawa**, hereby declare that the deep learning algorithms, telemetry ingestion pipelines, database schemas, and frontend architectures described in this technical dossier represent my original engineering work. Code repositories, commit logs, and production demonstrations are available upon request for technical verification and live code walk-throughs during recruiter technical interviews.

VIII. ACADEMIC & TECHNICAL REFERENCES

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